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RESEARCH REPORT

Presenting time-series data as absolute versus relative changes impacts judgments and choices

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Abstract

We examined how different ways of presenting time-series data affect consumer judgments and behaviors. Specifically, we compared the use of absolute terms (e.g., actual dollar change) versus relative terms (e.g., annual percentage change) in charts. Our findings revealed that absolute charts tend to have a more positive visual slope than relative charts, leading consumers to extrapolate these trends and expect faster growth in quantities. In Study 1, participants who viewed COVID-19 data in absolute charts, as opposed to relative charts, donated more money to a COVID-19-related charity and were less inclined to challenge quarantine advisories throughout the day. In Study 2, we observed that participants preferred investing in a country when its GDP was presented in an absolute chart, and visual trends were found to drive this effect. Study 3 demonstrated that individuals placed greater emphasis on absolute charts depicting population growth, particularly when the unit of measurement was more meaningful. Overall, the choice to represent data in absolute versus relative terms has a far reaching impact on judgments and choices.

KEYWORDS

communication, consumer behavior, COVID-19, data communication, information processing, investment decisions, judgment and decision making

INTRODUCTION

Clear and objective time-series data are essential for understanding trends such as disease spread, inflation, and population growth. However, these changes are often shown using one of two chart types: *absolute* changes in the units of interest or *relative* changes of these units in reference to their running stock.

Consider COVID-19 data, which can be expressed in both absolute (i.e., changes in the number of actual cases) and relative terms (i.e., percentage changes in the number of cases). Various news outlets, such as The Washington Post, Reuters, and The New York Times, have used absolute charts to represent the data, while The Atlantic, Financial Times, and Mother Jones have used relative charts (Figure 1). Similarly, both absolute and relative charts are used to track population changes (Frey, 2021: both absolute and relative), GDP growth (The World Bank, 2022: both absolute and relative), inflation (Bureau of Labor Statistics, 2023: relative; Federal Reserve Bank of St. Louis, 2023: both absolute and relative), and global warming metrics (Global Carbon Project, 2022: relative; NASA, 2023: absolute).

These two chart types frequently exhibit divergent visual trends, despite their shared underlying data. We find that because of this, a communicator's choice of chart type can impact people's judgments and behaviors. For example, consider the growth in iPhone users over time. Although the absolute growth in users is visually rising (Figure 2a), the same data presented in relative terms show a downward trend (Figure 2b). This happens because relative changes are calculated on a

Both authors contributed equally to this research

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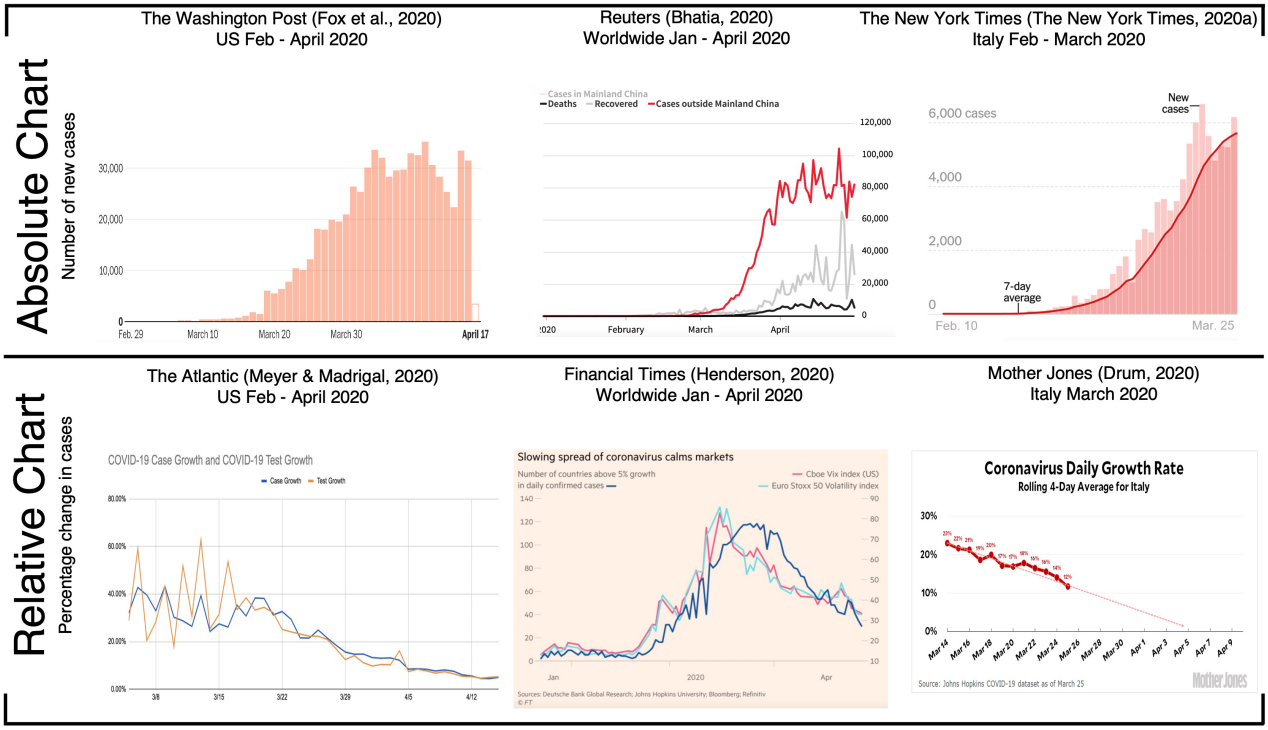

FIGURE 1 Daily changes in COVID-19 rates represented in absolute charts versus relative charts as reported in six different news articles.

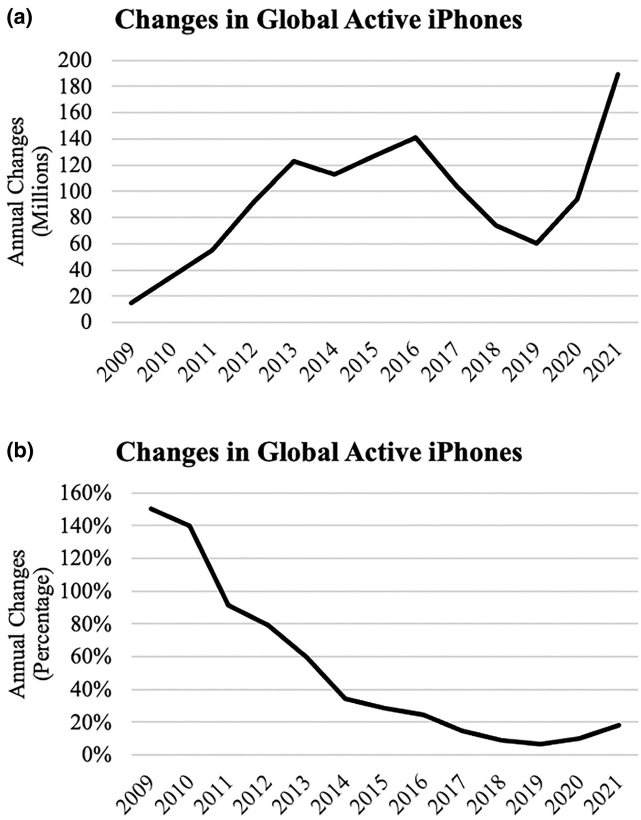

FIGURE 2 Annual changes in global active iPhone units. Panel (a) Absolute chart (2009–2021). Panel (b) Relative chart (2009–2021). *Source*: https://www.statista.com/statistics/1337745/active-apple-iphone-units-worldwide/.

growing denominator at each data point, unlike absolute changes.

We predict that when a person views an absolute versus relative chart, they will tend to expect the underlying quantity to grow more in the future. This is because absolute charts tend to have more positive visual slopes than relative charts. Past research has shown that people tend to extrapolate trends (Chapman, 2000; Guyse et al., 2002), which we believe will also occur with time-series data in absolute and relative charts. Recent research has also examined the visual differences between absolute charts in either stocks, representing a total quantity of a unit at each time point, or flows, representing changes in these quantities at each time point (Reinholtz et al., 2021; Spiller et al., 2020). These authors find that presenting changes in stocks versus flows can result in visually distinct patterns and corresponding differences in judgments. In contrast, we look exclusively at flows and distinguish between the visual differences of their absolute and relative presentation formats, and how these differences impact real and hypothetical judgments and behaviors.

Given the consequences of these differential interpretations across chart types, we also study when these judgment differences will be most pronounced. Specifically, we investigate the role of unit meaningfulness as a factor that moderates when consumers will pay more attention to an absolute versus relative chart. We define a unit as "more meaningful" when it is more directly tied to a real, underlying quantity and "less meaningful" when a unit

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abstractly represents an underlying quantity. For example, we would consider "people" in a population as a more meaningful unit than a "population-level index."

We predict that when a unit is more meaningful, that is, a closer reflection of an actual quantity, people will focus more on an absolute chart than on a relative one. We predict this will occur because consumers are more able to interpret meaningful units in an absolute versus relative chart. Absolute charts are able to represent real units, such as people, whereas relative charts are, by construction, less meaningful because a percentage change metric already represents an abstraction of an underlying quantity. Thus, we propose that when a unit of measurement is more meaningful, then people will pay more attention to the absolute versus relative chart, since only the absolute chart can show the real underlying quantity. This is consistent with prior literature which finds that people afford more attention to meaningful as compared to abstract units (Hoffrage et al., 2000; Lembregts & Van Den Bergh, 2019). However, when the unit of measurement becomes less versus more meaningful, then some share of people's attention will shift from the absolute toward the relative chart since both charts will then be representing an abstraction of an underlying unit.

To investigate differences in judgments and choices across absolute versus relative charts, we conducted eight preregistered experiments, three of which are included in the manuscript, with the other five included in the Appendix S1. In Study 1, we presented a statistic-savvy sample with real COVID-19 data and then followed up 2 days later, finding that they were more likely to have stayed in their home during the intervening day when they saw an absolute chart (with a visually upward slope) versus a relative chart (with a downward slope). In Study 2, we found participants preferred to invest in a country whose GDP was represented by an absolute compared to a relative chart, partially due to visual differences across the chart types. We also found that this effect is not influenced by mathematical ability. In Study 3, we examined our effect in a population growth domain and found that people rely more on absolute compared to relative charts when the charts convey more meaningful units. In Appendix S1, Supplemental Studies 1, 2 and 3 are conceptual replications of Study 1, while Supplemental Studies 4 and 5 explore the impacts of chart types in investment domains. All study preregistrations, data, do-files, and materials can be found at the following link: https://osf.io/wae8q/?view_only=5053be50665442e884a7b26f6a1042be.

STUDY 1

In Study 1, we presented a sample of educated participants with real-time COVID-19 data in either an absolute or relative chart and followed up 2 days later to record whether they left their house during the intervening day. We predicted that participants who viewed the absolute versus relative chart would be less likely to leave their homes and would donate more to a COVID-19-related charity.

Methodology

We recruited a U.S. sample ($N=523$, $M_{age}=34.88$, SD = 11.68; 47.8% female) with at least a bachelor's degree on Prolific Academic for £0.50 in the first round and £0.21 in the second round. As participants were American, these payments were advertised in USD.

This study took place in May of 2020, a time by which most states had issued stay-at-home orders (The New York Times, 2020b). Thus, exiting the house meant challenging quarantine advisories for most participants.

In the first round, we recruited 609 participants and randomly assigned them to view real, up-to-date, COVID-19 data in an absolute (Figure 3a) or relative (Figure 3b) chart. Next, as a comprehension check, we asked participants whether they thought the y-axis of the chart they viewed represented an absolute or relative change. We then told them that they were entitled to a $0.10 bonus but that they could donate any proportion of this they saw fit to Project C.U.R.E., a charity helping hospitals handle COVID-19. After indicating the amount that they wanted to donate, participants took a brief statistical ability quiz (DelMas et al., 2007) and answered a series of background questions.

Two days later, we invited participants to return for the second survey. We note that from the participants' perspective, the first and second rounds were conducted by different experimenters pursuing different projects, so participants would have no knowledge that the two surveys were connected.

For the second round, 523 participants returned with attrition not differing across conditions ($\chi^2(1, N=597)=0.0001$, $p=0.993$). Some participants gave unusable Prolific IDs in the first round, making it impossible to invite them to the second round, or participated in at least one round of the study twice. We excluded all of these participants from analyses ($n=12$). In this final sample, participants had at median a bachelor's degree, a median annual household income of $50–75K, consumed a median 30–60 min of COVID-19 news per day, and correctly answered a median 3 out of 4 advanced statistical questions from the ARTIST test of statistical comprehension (DelMas et al., 2007).

We asked participants whether they left their home during the intervening day (i.e., the day after the first round) in a yes or no response format. Participants then reported their expertise in statistics and were thanked, debriefed, and paid.

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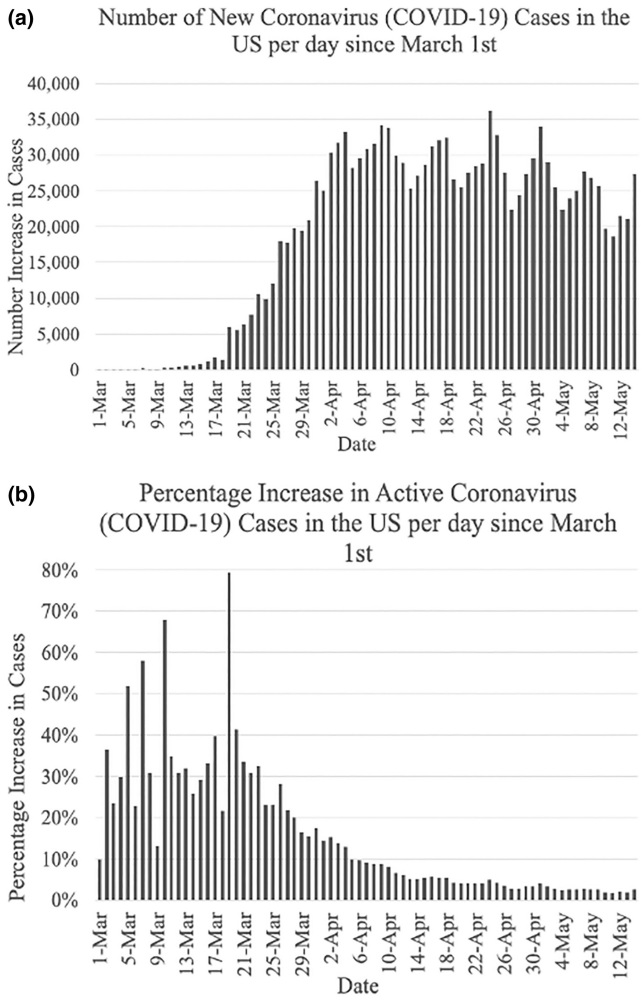

FIGURE 3 Study 1 absolute (a) and relative (b) charts. Panel (a): Absolute chart (March 1st, 2020–May 12th, 2020). Panel b: Relative chart (March 1st, 2020–May 12th, 2020). The data used for all studies using COVID-19 data were accessed through the Humanitarian Data Exchange and were compiled by the Johns Hopkins University Center for Systems Science and Engineering (JHU CCSE): https://data.humdata.org/dataset/novel-coronavirus-2019-ncov-cases. In Study 1 and Supplemental Study 1, we constructed the relative charts with *active* cases in the denominator, while in Supplemental Studies 2 and 3, we constructed the relative charts with *total* cases as the denominator. We varied active versus total cases as the denominator across studies to ensure our findings are robust to both operationalizations.

Results

Indicating that most participants understood the charts, those who viewed the absolute chart tended to correctly identify that their chart reflected absolute changes (92.5%), and those who viewed the relative chart tended to correctly identify that their chart reflected percentage changes (95.4%), with no differences in correctness across conditions ($\chi^2(1, N=523)=1.97$, $p=0.161$, $w=0.06$).

Consistent with our primary predictions, fewer participants left their homes the day after viewing the absolute (56.3%) versus relative chart (67.5%; $\chi^2(1, N=523)=6.99$, $p=0.008$, $w=0.12$). Further, participants donated more of their bonus to COVID-19 relief after viewing the absolute ($M=\$0.06$, SD=\$0.04) versus relative chart ($M=\$0.05$, SD=\$0.04; $t(521)=2.63$, $p=0.009$, $d=0.23$). These results also held when excluding observations from participants that failed the comprehension check (Appendix S1: A). We also found no evidence to suggest that statistical knowledge influenced the effects of relative versus absolute charts on participants' decisions to leave their homes, even after accounting for performance on the statistical knowledge quiz, self-reported statistical proficiency, and participants' educational background (Appendix S1: A).

Discussion

Viewing a COVID-19 chart for a brief period affected real self-distancing behaviors. Notably, a sample of educated participants left their homes less frequently after viewing COVID-19 cases in an absolute versus relative chart. Table 1 provides 5 additional replications of these findings.

STUDY 2

In Study 2, we investigated whether the differences in judgments about absolute and relative charts established in Study 1 can be explained by their distinct visual appearances. We did this by manipulating the visual distinctiveness between absolute and relative charts showing changes in a country's GDP. We predicted that if visual differences across chart types play a role in our effect, then reducing these differences should attenuate our effect.

We showed participants the same GDP changes in an absolute chart (with an upward visual trend) or a relative chart (with a downward visual trend). However, we also manipulated the extremity of the visual differences across these charts by changing the height of the y-axis. Thus, the study was a 2 (absolute vs. relative) × 2 (prominent vs. subtle visual trend) between subject design. We expected that people would think the country's economy would grow more if they saw the absolute chart than if they saw the relative chart but also predicted that this difference would be attenuated in the subtle versus prominent visual trend condition.

Additionally, in Study 2 we directly examine the role participants' mathematical abilities play in how they interpret absolute versus relative charts. We included a task where participants directly convert between chart types, allowing us to test whether the differences in chart interpretations in our previous studies vary as a function of these abilities.

Methodology

For Study 2, we recruited a U.S. sample through Prolific Academic ($N=600$, $M_{age}=38.95$, SD=13.11;

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TABLE 1 Supplemental studies' overview.

Study	Variation from study 1	Findings
Supplemental Study 1 (N=528; Pollfish) (Appendix S1: B)	We measured the within-participant changes in the number of hours they spent outside of their homes the day after viewing COVID-19 cases in an absolute versus a relative chart	Participants that saw the absolute versus relative chart were less likely to challenge quarantine advisories (p=0.002, d=0.27), indicated that they perceived the trend in COVID-19 cases as more toward increasing (p<0.001, d=0.76), and thought the threat of the disease had peaked later (p<0.001, d=2.55)
Supplemental Study 2 (N=523; Pollfish) (Appendix S1: C)	Participants viewed an absolute versus relative chart as in Study 1, then allocated a real $0.10 donation between a COVID-19 or world hunger-related charity and assessed other behavioral questions	Participants that saw an absolute versus relative chart donated more to the COVID-19-related charity (p=0.002, d=0.27), were less likely to attend a social gathering the next day (p<0.001, w=0.17), were more in favor of tightening restrictions (p=0.002, d=0.27), and thought that disease cases were increasing at a greater rate (p<0.001, d=0.44)
Supplemental Study 3 (N=501; Prolific) (Appendix S1: D)	Participants viewed absolute versus relative COVID-19 charts from Brazil, Canada, India, Mexico, Russia, or U.S. and responded to a series of behavioral questions	Participants that saw an absolute versus relative chart were less willing to leave their house and socialize (p<0.001, d=0.40), were more in favor of tightening restrictions (p<0.001, d=0.53), and thought that disease cases were increasing at a greater rate (p<0.001, d=1.11)
Supplemental Study 4 (N=298; Prolific) (Appendix S1: E)	Participants considered two investment advisors with identical past performances, but with these performances presented in an absolute (visually upward trending) or relative (visually downward trending) chart	The majority of participants preferred to invest with the advisor whose performance was shown in absolute versus relative terms (72.1%; z=3.82, p<0.001)
Supplemental Study 5 (N=200; Prolific) (Appendix S1: F)	Participants considered a share's price performance represented as an absolute versus relative *stock*, rather than as *flows* like in our other studies	In the absolute condition, 90.7% of participants thought that the stock was performing better in recent years, whereas only 53.0% of participants in the relative condition held this view (p<0.001, w=0.42)

48.8% female) and compensated them £0.40 for their participation. We excluded 23 participants from the sample due to their inability to identify the value of a U.S. penny.

We began by telling participants that they would be viewing changes in a country's GDP from 1960 onward, with the country's GDP in 1959 being $500 billion. We then showed them these changes in either absolute or relative terms, varying whether the visual trend was prominent (Figure 4a,b) or subtle (Figure 4c,d). Of note, the vertical axis was approximately 16 times larger in the prominent condition compared to the subtle condition, making the visual trend appear to have a steeper slope. All other aspects of the charts were identical, and participants were provided with the average growth change in both conditions.

As dependent measures, we asked participants the extent to which they expected this economy to grow versus shrink in the next 5 years (1 = Will majorly shrink, 7 = Will majorly grow) and the extent to which they would be willing to invest their savings in an index fund which provided returns based on this country's GDP (1 = Would not at all be willing to invest, 7 = Would be very willing to invest).

Additionally, we tested participants' abilities to convert between these two chart types. Similar to the study's main vignette, we asked them to consider a country's growth in GDP since 1959. We then showed them a series of three absolute and three relative charts. After each chart, participants had to indicate which of two other charts of the alternative type represented the same changes (see Appendix S2: Study 2 Materials).

Lastly, we thanked, debriefed, and paid participants.

Results

First, we conducted a 2 (absolute vs. relative) × 2 (prominent vs. subtle visual trend) ANOVA with anticipated economic growth as the dependent variable, finding a significant interaction (F(1, 573)=6.25; p=0.013, η^2=0.01). Planned contrasts revealed that among participants who saw prominent trends, those who saw absolute charts estimated more growth (M=4.97, SD=1.10) than those who saw relative charts (M=3.07, SD=1.30; t=13.99, p<0.001, d=1.57). We also found that among participants who saw subtle trends, those who saw absolute charts estimated more growth (M=4.73, SD=1.01) than those who saw relative charts (M=3.31, SD=1.09; t=10.81, p<0.001, d=1.36).

Next, we conducted a 2 (absolute vs. relative) × 2 (prominent vs. subtle visual trend) ANOVA with willingness to invest as the dependent variable, finding a significant interaction (F(1, 573)=10.18; p=0.002, η^2=0.02). Planned contrasts revealed that among participants who saw prominent trends, those who saw absolute charts were more willing to invest (M=4.54,

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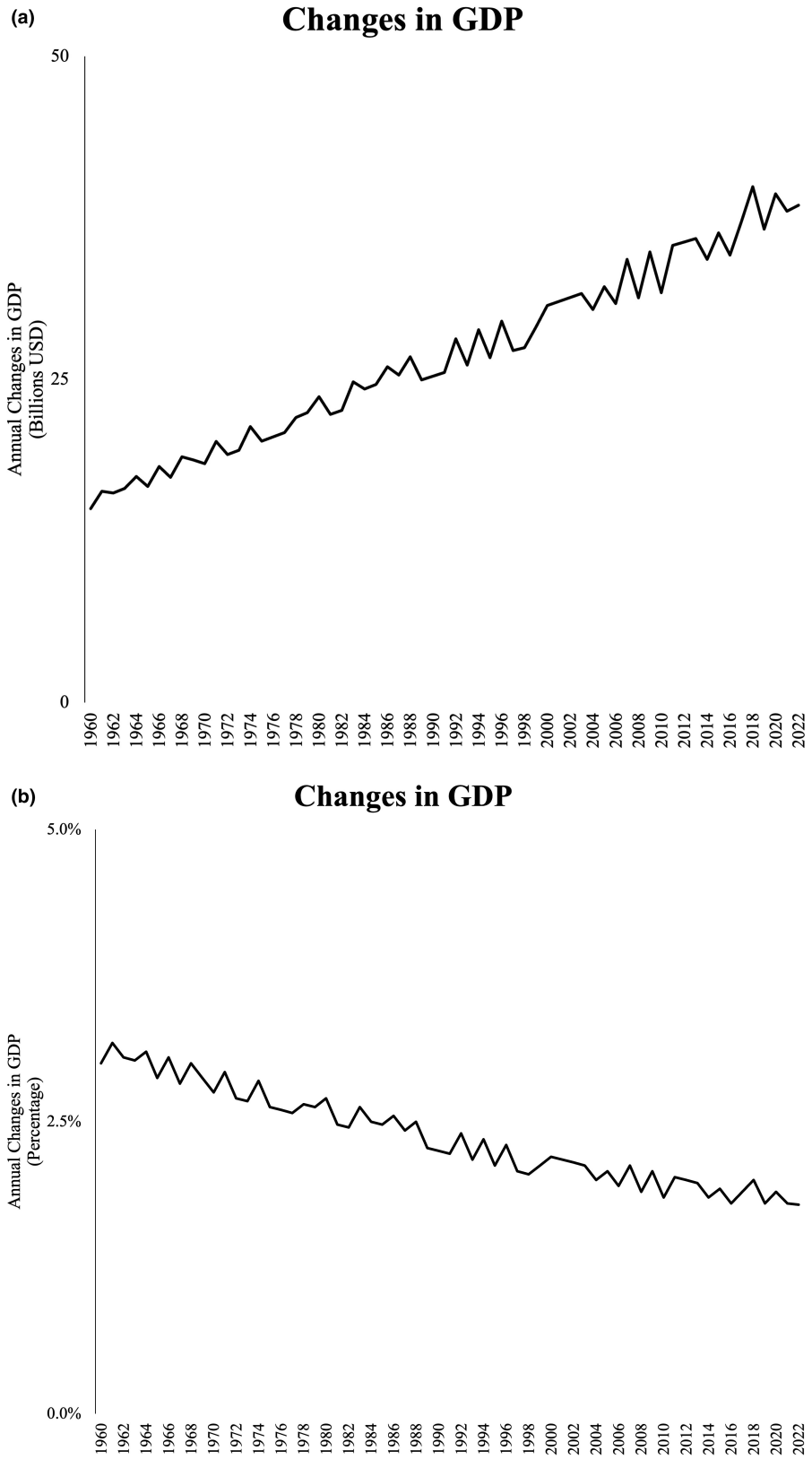

FIGURE 4 Study 2 charts. Panel (a) Absolute chart, prominent visual trend. Panel (b) Relative chart, prominent visual trend. Panel (c) Absolute chart, subtle visual trend. Panel (d) Relative chart, subtle visual trend.

SD = 1.62) than those who saw relative charts ($M = 2.14$, SD = 1.42; $t = 13.17$, $p < 0.001$, $d = 1.58$). We also found that among participants who saw subtle trends, those who saw absolute charts were more willing to invest ($M = 4.11$, SD = 1.58) than those who saw relative charts ($M = 2.52$, SD = 1.44; $t = 8.98$, $p < 0.001$, $d = 1.04$).

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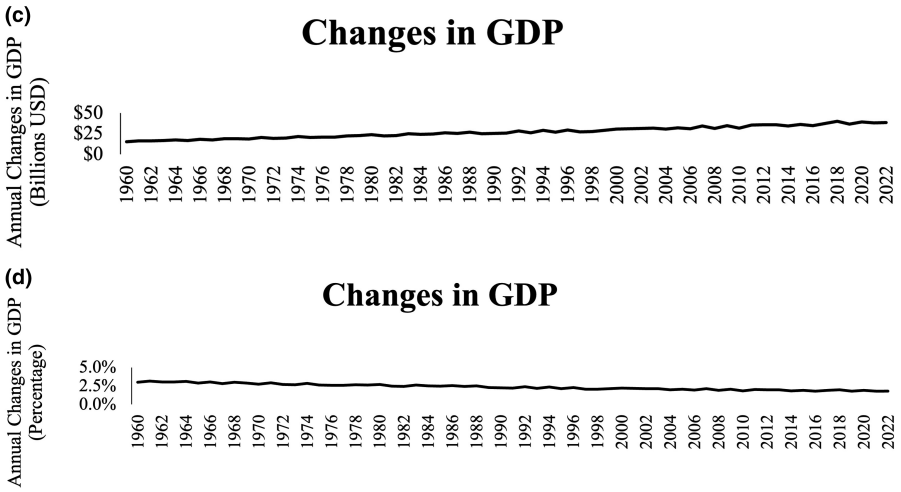

FIGURE 4 (Continued)

We then conducted these two same ANOVAs but added two terms: one for the number of correct answers participants had when converting between chart types (M=4.44, SD=1.27) and another for this measure's interaction with whether the chart they viewed was absolute or relative. After adding these terms, we found similar results for our key interactions with both dependent variables. Furthermore, we conducted our main analyses only with participants that got at least five of the six conversion questions correct (n=339), finding our results still held. These analyses can be found in Appendix S1: G.

Discussion

In Study 2, we found that if absolute and relative charts look more visually similar, people's corresponding judgments of them are likewise more similar. This shows that a chart's visual slope influences how people make decisions. Potentially because there is still a present visual difference across these charts, we observed partial, rather than full attenuations of our effect across our two measures (i.e., the effect was still present in the subtle visual trend conditions). In the extreme, a presenter could reduce the y-axes of both charts to the point of any visual trends being indistinguishable. However, while this may eliminate trend-based differences in judgments across chart types, it would also render these charts virtually uninterpretable. Furthermore, we found that participants' abilities to convert between chart types did not play a role in their interpretations of these charts. Thus, our effect seems to be driven by differences in the charts themselves, rather than by ignorance toward the relationship between these chart types.

STUDY 3

In Study 3, we explore how unit meaningfulness moderates the attention given to different chart types. Participants saw variations in a city's population using both absolute and relative charts. Notably, the absolute chart demonstrated an upward visual trend, while the relative chart depicted a downward visual trend. We distinguished between unit meaningfulness by displaying changes in an actual population (labeled as "more meaningful"' because of its direct tie to a real, tangible underlying quantity) versus a population-level index (labeled as "less meaningful" owing to its abstract representation of an underlying quantity). We hypothesized that when the charts reflect changes in the more real unit of an actual population, as contrasted with the abstract "population index," individuals would perceive the population as growing more rapidly. We predicted that this perception would stem from individuals placing greater attention on the absolute chart when presented with more meaningful units.

Methodology

For Study 3, we recruited a U.S. sample through Prolific Academic (N=299, M_{age}=38.37, SD=13.23; 49.8% female) and compensated them £0.20 for their participation. Fifteen participants were excluded from the sample due to their inability to identify the value of a U.S. penny.

We began by telling participants that we would be showing them two charts, documenting the changes in a city's population since 1950. These charts were shown in a random order and displayed absolute and relative changes in the city's population. In a population condition, we presented these charts as showing changes in the city's actual population, whereas in an index condition, we labeled the same charts as showing changes in points of a city's population index. To convert the absolute chart from the population to index condition, we relabeled the y-axis as representing changes in points, rather than changes in thousands of people. The absolute charts showed visually upward trends, while the relative charts displayed visually downward trends.

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As our dependent measure, we asked participants to estimate the extent to which the city's population will grow or shrink over the next 10 years (−3 = Will significantly shrink, 0 = No change, 3 = Will significantly grow).

As comprehension checks, we asked participants whether the charts they viewed represented changes in the actual population or in a population index, whether the absolute chart represented absolute or relative changes in the population (or population index), and whether the relative chart represented absolute or relative changes in the population (or population index).

As an exploratory measure of attention, we then asked participants the extent to which they relied on the absolute versus relative chart when estimating population growth for this city (−3 = Completely relied on the relative chart, 0 = Relied evenly on both charts, 3 = Completely relied on the absolute chart).

Lastly, we thanked, debriefed, and paid participants.

Results

Consistent with our prediction, participants thought the city's population would grow more in the population ($M = 0.79$, SD = 1.41) versus index condition ($M = 0.13$, SD = 1.65; $t(283) = 3.65$, $p < 0.001$, $d = 0.43$). Additionally, we found that participants relied more on the absolute compared to the relative chart in the population ($M = 0.59$, SD = 1.62) versus index condition ($M = 0.12$, SD = 1.61; $t(283) = 2.44$, $p = 0.016$, $d = 0.29$).

We then tested whether the effect of a unit meaningfulness on change estimates was mediated by a participant's reliance on the absolute versus relative chart. To do this, we first created a dummy variable which equaled 0 if a participant saw the population index and 1 if they saw the actual changes in the population. We then estimated an indirect effect using bootstrapped standard errors based on 10,000 resamples. We found a significant indirect effect of unit type on change estimates via the extent to which a participant relied on the absolute versus relative chart ($b = 0.22$, 95% CI = [0.04, 0.41], $p = 0.018$).

All of these analyses held after excluding observations from participants that failed the comprehension checks ($n = 52$; Appendix S1: H).

Discussion

In Study 3, we found evidence that unit meaningfulness moderates the attention given to different chart types. When units were more meaningful, participants were more influenced by the absolute chart; with less meaningful units, they leaned more toward the relative chart for population growth assessments.

GENERAL DISCUSSION

The current research shows that representing charts in absolute versus relative terms significantly influences judgments and behaviors. We observed that the type of chart affected judgments across multiple domains, such as the spread of COVID-19 (Study 1 and Supplemental Studies 1–3), a country's economic growth (Study 2), population growth (Study 3), investment-advisor performance (Supplemental Study 4), and a company's share price (Supplemental Study 5). These effects remained consistent even after accounting for statistical (Study 1) and mathematical skills (Study 2) and seemed to be primarily driven by visual trends (Study 2).

Limitations and future directions

In Study 2, we discovered that participants interpret identical data differently depending on the chart type due to perceived visual trends. However, these visual differences did not fully account for the varied chart interpretations, suggesting multiple mechanisms may be at play. For example, individuals often display an exponential growth bias, failing to recognize the increasing stock of a quantity when assessing relative changes (Almenberg & Gerdes, 2012). Such a bias might cause misinterpretations of relative charts, as individuals might overlook compounded growth. Broadly, when deciphering these charts, individuals likely utilize a mix of various heuristics (Hutchinson et al., 2010).

Moreover, given that visual trends are at least partly responsible for this effect, it suggests significant potential boundary conditions. Specifically, there could be instances where both absolute and relative charts display the same visual trend, as seen in cases of exponential growth. Although the magnitude of their slopes might differ, the similarity in direction could result in comparable trend interpretations, possibly diminishing our observed effect.

Furthermore, people probably depend on these visual trends partly because they do not engage in the mental computation required to transition between these chart types. While we suspect this is the prevalent scenario in real-world chart interpretations, future research exploring situations where this effect diminishes due to motivated mental conversions would be beneficial.

Lastly, in Study 3 we found that more meaningful units of change led to participants weighting absolute (vs. relative) charts more in their judgments. This finding builds on existing work which shows that more meaningful units are more easily interpreted (Hoffrage et al., 2000; Lembregts & Van Den Bergh, 2019). However, this raises the question of what other judgment consequences might arise from a unit's meaningfulness. For instance, changes conveyed in more meaningful units could be regarded as more relevant, since they usually represent more real

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units (e.g., global carbon emissions), rather than an abstraction (e.g., a climate-change threat index).

Public-facing institutions ought to present consumers carefully chosen data representations, particularly when such data influence behaviors and well-being. This is especially true as consumers are increasingly presented with data and figures they may not be trained to interpret. For instance, a great deal of research has shown consumers were strongly, and sometimes erroneously, influenced by data presentations in the COVID-19 pandemic when making health judgments (Harman et al., 2021; Padilla et al., 2022; Reinholtz et al., 2021; Romano et al., 2020; Ryan & Evers, 2020). Thus, relevant institutions should be careful in how they present chart data. Under the correct circumstances, they may even be able to use chart type as a nudge to improve consumer decision making. Understanding the impact of chart type on decision making is imperative when lives and well-being hang in the balance.

DATA AVAILABILITY STATEMENT

All data, analysis files, and pre-registrations are available at the following link: https://osf.io/wae8q/?view_only=5053be50665442e884a7b26f6a1042be.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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